

Diffusion of linear operators: the algorithm*

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Problem statement for the linear SCM

There given a collection of pairs $(\mathbf{D}, \mathbf{Z})$ with the dataset $\mathbf{D} = \{\mathbf{x}\}$. The elements of the vector $\mathbf{x}$ is generated by the random variables $X_1, \dots, X_n$ according to an unknown probabilistic model. The adjacency matrix $\mathbf{Z}$ corresponds to the graph $G(V, E)$ with the nodes $V = \{X_1, \dots, X_n\}$ and the edges $E = \{(X_i, X_j) \mid Z_{ij} = 1\}$.

Fix the structure of a model as a linear operator $\mathbf{W}$ that maps a sample $\mathbf{x}$ to its reconstruction $\hat{\mathbf{x}}$,

$$\mathbf{0} = \mathbf{W}^\top \mathbf{x}.$$

The elements of the adjacency matrix $\mathbf{Z}$ is the $n \times n$ binary matrix that corresponds to the elements of $\mathbf{W}$, the $n \times n$ matrix of the linear operator.

This matrix is generated from the normal distribution¹ subject to its columns

$$\mathbf{W} = [\mathbf{w}_1 \quad \mathbf{w}_2 \quad \dots \quad \mathbf{w}_n], \quad \text{where} \quad \mathbf{w} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{A}).$$

The following assumptions are made:

1. The dot product $\mathbf{w}^\top \mathbf{A} \mathbf{x}$ define a Riemannian metric on the space of the model parameters $\mathbf{w}$. The non-singular PSD matrix $\mathbf{A}$ is a metric tensor.

*Simplest version. Without the part where the generative AI learning procedure. And with reduced do-intervention procedure.

¹Generalized Linear Models by J. A. Nelder and R. W. M. Wedderburn, 1972.

2. The matrix $\mathbf{W}$ is sparse because the corresponding graph G is sparse. So the item above should be statistically tested.
3. The representation of the SCM model in the form

$$\mathbf{0} = \mathbf{W}^\top \underline{\mathbf{A}} \mathbf{x},$$

where $\underline{\mathbf{A}}$ is a tree-way tensor for a quadratic form $\mathbf{w}^\top \underline{\mathbf{A}} \mathbf{x}$ delivers a better quality of the causality reconstruction. In the indexed notations it is $\varepsilon_k = \sum_{i,j} w_{ik} A_{ijk} x_j$ with $i, j, k = 1, \dots, n$.

4. Along with the dataset $\mathbf{D}$ and the corresponding causal graph $\mathbf{Z}$ we have a prior distribution $p(\mathbf{Z} \mid \mathbf{D})$.

Formulate the reconstruction problem. Find the matrix $\mathbf{Z}$ along with the matrix of the optimal parameters $\hat{\mathbf{W}}$ and its covariance matrices $\underline{\mathbf{A}}$ so that the dataset $\mathbf{D}$ brings the (approximate) kernel to the linear system

$$\ker(\mathbf{W}) = \{\mathbf{D} \ni \mathbf{x} : \|\mathbf{W}^\top \mathbf{x}\|_2^2 \rightarrow \min\}.$$

Note that the kernel of the linear operator $\mathbf{W}$ is a subspace of the observable data $\mathbf{D}$. Note the singularity of the matrix $\mathbf{W}$ is a necessary condition for the existence of the kernel. Note that singular values decomposition of the matrix $\mathbf{W}$

$$\mathbf{W} = \mathbf{U} \mathbf{\Lambda} \mathbf{V}^\top$$

provides a basis of the kernel. Denote by $d = \dim(\mathbf{W})$.

The diffusion algorithm

The algorithm of the causal graph reconstruction.

To control the diffusion process use a rejection sampling algorithm. As a rejection sampling, it comprises three components:

- 1) sampling,
- 2) computing the criterion, and
- 3) acceptance or rejection.

Below is a genetic version of this algorithm.

Get the pair $(\mathbf{D}, \mathbf{Z})$ from the collection and set them as $(\mathbf{D}_0, \mathbf{Z}_0)$. If there is no prior distribution $p(\mathbf{Z} \mid \mathbf{D})$, then set $\mathbf{Z}_0$ as a random graph. Set the time $t = 0$ and run for $t = 0, \dots, T$.

1. Sampling

1. **Inheritance.** Sample $\mathbf{Z}_{t+1} \sim p(\mathbf{Z}_{t+1} \mid \mathbf{Z}_t)$: add a random parent to each child with a small probability π_1 .
2. **Ground truth.** If there is a prior, then sample $\mathbf{Z}_{t+1} \sim p(\mathbf{Z}_0)$: set 1 or 0 according to the prior $\mathbf{Z}_0$ with a small probability π_2 .
3. **Testing.** Sample according to the do-intervention procedure: set 1 if the corresponding element $\delta\alpha_{ij}$ in the do-pair $(\mathbf{A}, \mathbf{A}')$ is statistically significant with a small probability π_3 .
4. **Complexity.** Sample according to the model selection criteria: invert random elements of the matrix $\mathbf{Z}$ with a small probability π_4 . The complexity distribution $p(\mathbf{Z} \mid t, d, h)$ depends on the time step t , the dimensionality the operator d_t , and the entropy of diffusion h .

2. Computing the Criterion

The composite function for the rejection sampling

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\mathbf{D}} + \lambda_2 \mathcal{L}_{\mathbf{W}} + \lambda_3 \mathcal{L}_C, \quad \sum \lambda_i = 1$$

includes three components:

1. **Data loss:**

$$\mathcal{L}_{\mathbf{D}} = (\hat{\mathbf{x}} - \mathbf{x})^\top \mathbf{B} (\hat{\mathbf{x}} - \mathbf{x}),$$

where the binary matrix $\mathbf{B}$ is a diagonal and its elements are zero for the children without parents in $\mathbf{Z}$ (this includes unused nodes X).

2. **Statistical complexity:** the norm of elementwise product

$$\mathcal{L}_{\mathbf{W}} = \|\mathbf{W} \odot \mathbf{A}\|_2^2.$$

a variant

$$\mathcal{L}_{\mathbf{W}} = \|(\hat{\mathbf{W}} - \mathbf{W})^\top \underline{\mathbf{A}} (\hat{\mathbf{W}} - \mathbf{W})\|_2^2.$$

3. **Structural complexity:** dimensionality of $\mathcal{L}_C = d$, the last d -th non-zero value of the singular values decomposition of the matrix $\mathbf{W}$.

The criterion computing procedure:

1. For the sampled $\mathbf{Z}_t$ and the data $\mathbf{D}$ compute the parameters $\mathbf{W}$ to minimize the loss $\mathcal{L}$. Set the covariance matrix $\mathbf{A} = \mathbf{I}$ at $t = 0$. *Note that there is an analytic way to compute the parameters $\mathbf{w}$ and Bayesian hyperparameters $\mathbf{A}$ for the linear regression models $\mathbf{w}^\top \mathbf{x}$.*
2. Fix the parameters $\mathbf{W}$ and estimate the covariance $\mathbf{A}$ for each parameter as independent variables. (As a variant, estimate the tensor $\underline{\mathbf{A}}$ to capture the dependencies between parameters.)
3. Sample the do-intervention dataset $\text{do}(\mathbf{D})$ and compute the new parameters $\mathbf{W}'$ and their covariance $\mathbf{A}'$. (To sample the do-intervention dataset, select a small random subset of the variables $X_1, \dots, X_n$ and set them to zero.)

3. Acceptance or Rejection

Instead a threshold rejection, the genetic algorithm keeps the best samples for further diffusion.

1. Sample N candidates $\mathbf{Z}_{t+1}$ and **Accept** the best K according to the composite criterion $\mathcal{L}$.
2. **Reject** the rest.

Note that there is to inter-population operations (like crossover) in this version of the algorithm.